

# Dataset Description

## 1. MovieLens Dataset

- Name / Description

The MovieLens dataset is a widely used movie recommendation dataset developed by the GroupLens Research Lab at the University of Minnesota. It contains information about movies, user ratings, and user-generated tags. In this project, we used the movies.csv, ratings.csv, tags.csv, and links.csv files.

- Source

URL: <https://grouplens.org/datasets/movielens/>

- Value to the Research Question

The MovieLens dataset provides user rating data and tagging activity for movies. These data allow us to calculate the average MovieLens rating per movie and calculate the average MovieLens rating per movie and measure user engagement through tag counts. These values are used to compute the disagreement between MovieLens ratings and ratings from other platforms, such as IMDb.

## 2. IMDb Dataset

- Name / Description

The IMDb dataset is an open dataset provided by IMDb containing extensive information about movies, including titles, release years, genres, ratings, and vote counts. In this project, we used title.basics.tsv and title.ratings.tsv.

- Source

URL: <https://datasets.imdbws.com/>

- Value to the Research Question

The IMDb dataset provides the IMDb average rating and number of votes for each movie. These metrics allow us to compare IMDb ratings with MovieLens ratings for the same movies. The IMDb average rating is used together with the MovieLens average rating to compute the alignment and disagreement metrics defined in our research question.

## 3. TMDB Dataset

- Name / Description

The TMDb dataset (The Movie Database) contains movie metadata such as titles, release dates, genres, and user ratings. In this project, we used the dataset `TMDb_movie_dataset_v11.csv`.

- Source

URL: <https://www.kaggle.com/datasets/asaniczka/tmdb-movies-dataset>

- Value to the Research Question

The TMDb dataset provides additional metadata about movies, including genre information and rating statistics. This dataset helps enrich the movie information and supports cross-platform comparison across MovieLens, IMDb, and TMDb.

## Data Cleaning Process

Before loading the datasets into the Oracle database, we performed several data cleaning and preparation steps using Python to ensure consistency across datasets.

### MovieLens Cleaning

For the MovieLens dataset, we retained only the columns necessary for linking the datasets and answering the research questions. From the `movies.csv` and `links.csv` files, we kept the attributes:

- `movieId`
- `title`
- `genres`
- `tconst` (IMDb identifier)
- `tmdbId`

The `links` table was used to connect MovieLens movies with IMDb and TMDb identifiers. No major missing values were found in the relevant columns.

For the **`ratings.csv`** file, we kept only the attributes `movieId` and `rating`, since the user identifier was not required for computing average ratings.

For the **`tags.csv`** file, we retained the attributes `movieId` and `tag`. The tag data is later used to calculate the number of tags per movie, which serves as a measure of user engagement.

### IMDb Cleaning

The IMDb dataset was filtered to retain only rows with valid identifiers and numeric values so that each movie could be reliably linked to the other datasets. This aligns with the scope of our research questions.

From the `title.basics.tsv` file, we retained the following attributes:

- `tconst`

- primaryTitle
- startYear
- genres

From the title.ratings.tsv file, we kept:

- tconst
- averageRating
- numVotes

Rows with missing identifiers or invalid numeric values were removed to ensure that all records could be reliably joined with the other datasets.

## **TMDB Cleaning**

The TMDB dataset was cleaned by converting the release\_date field into a date format and extracting the release year. Similar to the IMDb dataset, we filtered the dataset to include only movies released between 2020 and 2025.

Rows with missing imdb\_id values were removed because the IMDb identifier is required to link TMDB movies with the IMDb dataset.

Finally, we retained only the attributes relevant to our database schema and analysis:

- tmdb\_id
- imdb\_id
- tmdb\_title
- release\_year
- tmdb\_genres
- tmdb\_vote\_average
- tmdb\_vote\_count

## **Additional Subset Creation Due to Database Space Limitations**

During the process of inserting the cleaned datasets into the Oracle database, we encountered storage quota limitations on the department database server. The original cleaned datasets contained tens of thousands of rows, which exceeded the available storage space for our Oracle account.

To address this issue while still maintaining meaningful analysis, we created a matched project subset of the data. Instead of inserting the entire datasets, we selected a smaller subset of movies that could be consistently linked across all three platforms used in the project: MovieLens, IMDb, and TMDB.

First, we identified movies that appeared in all relevant datasets by using the linking identifiers provided in the MovieLens links table. This ensured that every movie in the subset could be connected to its corresponding records in the IMDb and TMDB datasets.

Next, we further filtered the data by release year and genre to keep a manageable subset that still represents a diverse set of films. From this filtered set, we randomly sampled 1000 movies to form the core dataset used in our relational database.

As a result, the following tables were reduced to approximately 1000 rows each:

- movies
- links
- imdb\_basics
- imdb\_ratings
- tmdb

For the ratings and tags tables, we retained all records associated with the selected 1000 movies rather than randomly truncating these tables. This approach ensures that important aggregations, such as average ratings and tag counts, can still be computed accurately using SQL queries within the database.

After creating the subset, the approximate sizes of the datasets became:

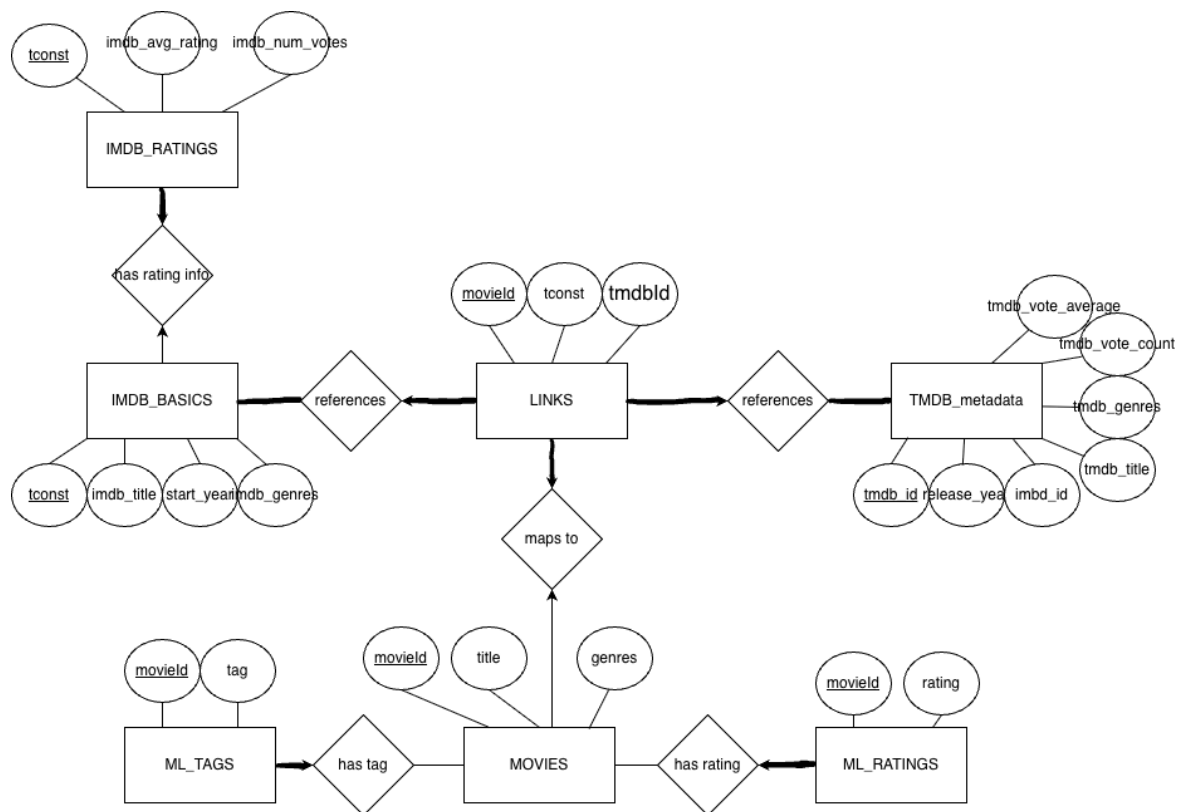
| Table        | Rows   |
|--------------|--------|
| movies       | ~1000  |
| links        | ~1000  |
| imdb_basics  | ~1000  |
| imdb_ratings | ~1000  |
| tmdb         | ~1000  |
| ratings      | ~13000 |
| tags         | ~3500  |

This subset preserves the relationships between datasets while keeping the data size manageable for insertion and querying within the Oracle database environment.

### Summary

- Overall, the cleaning process involved:
- A subset of 1000 movies was selected to satisfy the Oracle database storage quota.
- Aggregating MovieLens ratings to compute the average rating per movie
- removing rows with missing key identifiers
- retaining only attributes required for the analysis
- standardizing identifier formats to allow cross-dataset linking
- No joins or aggregations were performed during the cleaning stage, as these operations will be handled later using SQL queries within the Oracle database.

### ER Diagram



The ER diagram represents the relational structure used to integrate movie data from the MovieLens, IMDb, and TMDb datasets. The MOVIES entity serves as the central table containing the MovieLens movie identifier, title, and genre information. MovieLens ratings and tags are stored in the ML\_RATINGS and ML\_TAGS tables, which are linked to MOVIES through the movieid attribute.

The LINKS table maps MovieLens movie identifiers to external identifiers used by other datasets, specifically the IMDb identifier (tconst) and the TMDb identifier (tmdbid). These identifiers allow the database to connect MovieLens movies to additional metadata and rating information from external sources.

IMDb information is stored in two tables: IMDB\_BASIC, which contains general movie information such as title, release year, and genres, and IMDB\_RATINGS, which contains IMDb average ratings and vote counts. Both tables are connected through the tconst identifier.

Additional metadata from The Movie Database is stored in the TMDb\_METADATA table, which includes TMDb identifiers, genres, vote averages, and vote counts. The TMDb table is connected to the rest of the schema through the tmdbid attribute in the LINKS table.

Primary keys are defined for each entity (e.g., movieid for MOVIES and tconst for IMDb tables), while the LINKS table uses foreign keys to reference the corresponding identifiers in the external datasets.

This design allows the database to combine movie metadata and rating information from multiple platforms and supports cross-platform comparison of movie ratings.

## Overarching research question

The primary goal of this project is to examine how consistent movie ratings are across different rating platforms. In particular, we focus on comparing MovieLens ratings and IMDb ratings for the same movies that appear in both datasets. Since different platforms collect ratings from different groups of users and use different rating scales or voting systems, the ratings for the same movie may vary across platforms.

To quantify this difference, we compare the average MovieLens rating for each movie with the IMDb average rating. We define the alignment between the two platforms as the difference between these values:

$$\text{Alignment} = \text{IMDb\_avg} - \text{MovieLens\_avg}$$

A positive alignment value indicates that IMDb users tend to rate the movie higher than MovieLens users, while a negative value indicates that MovieLens users rate the movie higher.

To measure the magnitude of disagreement between platforms regardless of direction, we define disagreement as the absolute value of the alignment:

$$\text{Disagreement} = |\text{Alignment}|$$

Using these definitions, our analysis investigates how rating differences vary across movies and whether factors such as movie genre or user engagement influence the level of disagreement between platforms.

## Research questions:

**Question 1:** How consistent are movie ratings between IMDb and MovieLens for the same movies released between 2020 and 2025?

- **SQL Query:**

```
SELECT m.title,
       r.rating AS movielens_rating,
       ir.imdb_avg_rating,
       ABS(ir.imdb_avg_rating - r.rating) AS disagreement
FROM MOVIES m
JOIN ML_RATINGS r ON m.movieId = r.movieId
JOIN LINKS l ON m.movieId = l.movieId
```

```
JOIN IMDB_RATINGS ir ON l.tconst = ir.tconst

FETCH FIRST 10 ROWS ONLY;
```

**File:** phase3\_load\_data.ipynb

Location: Step 5 – Research queries section

- This query retrieves ratings for the same movies from both the MovieLens and IMDb datasets. The MOVIES table is joined with ML\_RATINGS, LINKS, and IMDB\_RATINGS using the movieId and tconst identifiers to match movies across datasets. The disagreement between the two platforms is calculated as the absolute difference between the IMDb average rating and the MovieLens user rating. This metric measures how consistent the ratings are across the two platforms.

- **Results:** Example output from the notebook:

| Title                              | MovieLens Rating | IMDb rating | Disagreement |
|------------------------------------|------------------|-------------|--------------|
| Eyimofe (This Is My Desire) (2020) | 3.66             | 7           | 3.33         |
| Break Even (2020)                  | 3.0              | 4.7         | 1.7          |
| Faith Under Fire (2020)            | 3.75             | 3.4         | 0.35         |
| The Family Tree (2020)             | 2.0              | 4.8         | 2.8          |

The results show that some movies have noticeable differences between MovieLens and IMDb ratings, indicating that user perception may vary across platforms.

**Question 2:** Does the average disagreement (mean |Alignment|) differ across selected genres (2020–2025)?

- **SQL Query:**

```
SELECT m.genres,
       AVG(ABS(ir.imdb_avg_rating - r.rating)) AS
avg_disagreement
FROM MOVIES m
JOIN ML_RATINGS r ON m.movieId = r.movieId
```

```

JOIN LINKS l ON m.movieId = l.movieId

JOIN IMDB_RATINGS ir ON l.tconst = ir.tconst

GROUP BY m.genres

ORDER BY avg_disagreement DESC

FETCH FIRST 10 ROWS ONLY;

```

**File:** phase3\_load\_data.ipynb

Location: Step 5 – Research queries section

- This query computes the average disagreement between IMDb ratings and MovieLens ratings for movies grouped by genre. The disagreement is calculated as the absolute difference between the IMDb average rating and the MovieLens rating. The query joins the MOVIES, ML\_RATINGS, LINKS, and IMDB\_RATINGS tables so that each movie has both rating values available. The results are grouped by genre, and the average disagreement is calculated for each genre, allowing comparison of rating differences across genres.
- **Results:** Example output from the notebook:

| Genre                  | Avg Disagreement |
|------------------------|------------------|
| Comedy Drama Thriller  | 5.23             |
| Action Horror Thriller | 4.8              |
| Action Comedy Thriller | 4.4              |
| Mystery                | 4.1              |

The results suggest that certain genres show larger disagreement between IMDb and MovieLens ratings, indicating that audience perception may vary depending on the type of movie.

**Question 3:** Is there a measurable correlation between user engagement (MovieLens tag counts and IMDb vote counts) and disagreement magnitude?

- **SQL Query:**

```

SELECT m.title,
       COUNT(DISTINCT t.tag) AS tag_count,

```



```

        ir.imdb_num_votes,

        ABS(ir.imdb_avg_rating - AVG(r.rating)) AS
disagreement

FROM MOVIES m

JOIN LINKS l ON m.movieId = l.movieId

JOIN IMDB_RATINGS ir ON l.tconst = ir.tconst

JOIN ML_RATINGS r ON m.movieId = r.movieId

LEFT JOIN ML_TAGS t ON m.movieId = t.movieId

GROUP BY m.movieId, m.title, ir.imdb_num_votes,
ir.imdb_avg_rating

HAVING COUNT(DISTINCT t.tag) > 0

ORDER BY disagreement ASC

FETCH FIRST 10 ROWS ONLY;

```

**File:** phase3\_load\_data.ipynb

Location: Step 5 – Research queries section

- This query examines whether user engagement is related to disagreement between IMDb and MovieLens ratings. User engagement is measured using the number of unique MovieLens tags for each movie and the number of IMDb votes. Disagreement is calculated as the absolute difference between the IMDb average rating and the average MovieLens rating for the same movie. The query joins the MOVIES, LINKS, IMDB\_RATINGS, ML\_RATINGS, and ML\_TAGS tables, groups the results by movie, and filters to movies with at least one tag.

- **Results:** Example output from the notebook:

| Title                         | Tag Count | IMDb Vote Count | Disagreement |
|-------------------------------|-----------|-----------------|--------------|
| Kitty Mamas (2020)            | 1         | 187             | 0.20         |
| Me You Madness (2021)         | 3         | 1589            | 0.76         |
| It's a Wonderful Binge (2022) | 1         | 1317            | 1.10         |

|                           |   |     |      |
|---------------------------|---|-----|------|
| The Vacationers<br>(2023) | 3 | 251 | 1.10 |
|---------------------------|---|-----|------|

The results show that movies with at least some tagging activity can still have a wide range of disagreement values. For example, some movies with low tag counts have small disagreement, while others with more tags or more IMDb votes have larger disagreement. This suggests that user engagement may be related to disagreement, but the relationship is not obviously linear from this small sample alone.

**AI Tool use declaration:** no AI tool was used for this assignment; all has been completed by ourselves.